

M/s, Zelle Biotechnology Pvt. Ltd.
21.03.2026

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Sr. No.	Change	Date	Approved By EPCE	Approved By Zelle

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25278-3520-0634

DS for Insulation Nitrile rubber

M/s, Zelle Biotechnology Pvt. Ltd.

Version-A

21.03.2026

Operating Data

Fluid	Chilled water, AHU drain			
	Minimum	Normal	Maximum	Unit
Pressure	2	3.5	6	kg/cm ² g
Temperature	6	12	50	°C
Atmospheric Conditions Considered	29.7 deg C and 87 % RH			

Specification of Insulation

Type	Closed Cell Elastomeric Nitrile Rubber			
Service	Pipe line			
Thermal Transmittance	*			
Thermal Conductivity	0.023 w/m° k @ temp 20 °C			
Density	40 ±2 Kg/m ³			
Moisture diffusion factor or μ	Minimum 7000			
Vapour permeability	*			
Fire performance	*			

Comply with deviation
Thermal conductivity for
nitrile rubber will be
0.036 w/mk at 20 C mean
temperature`

Insulation Thickness / *

For Chilled water @ 7 degC

Pipe Size (NB)	Insulation thickness with Aluminium cladding 24G (mm) External	Insulation thickness with GC Cloth (mm) Internal
15	19	38
20	19	38
25	25	44
32	25	44
40	25	44
50	32	50
65	32	50
80	32	57
100	38	57
125	38	64
150	38	64
200	44	70
250	50	70
300	50	70
350	50	70

Noted

For AHU drain @ 10 degC

Pipe Size (NB)	Insulation thickness with Aluminium cladding 24G (mm) External	Insulation thickness with GC Cloth (mm) Internal
15	19	19
20	19	19
25	19	25
32	19	25
40	19	25
50	25	25
65	25	25
80	25	25
100	32	25
125	32	32
150	32	32

Noted

Requirements

1. Insulation should have high stress corrosion inhibitors.

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|---|
| 2. The Insulated Pipe shall be Wrapped with 7 mill thk. black chemically treated woven glass cloth of 205 ± 5 GSM |
| 3. The insulation should be dry while installed & shall be kept dry. |
| 4. Full thermal efficiency shall restore on drying out. |
| 5. Resilient material to regain thickness & shape. |
| 6. The insulation shall have fire performance such that it passes Class 'O' as per BS 476 Part 6 & 7. |
| 7. Insulation shall be self extinguishing, shall not drip and spread the flame |

Mechanical Properties

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| 1. Dimensional stability, Compressive strength, Vibration resistant. |
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Notes

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| 1. All valves, fittings, joints, strainers, etc. in water piping shall be insulated to the same thickness as specified for the main run of piping. |
| 2. Valves bonnet, yokes and spindles shall be insulated in such a manner as not to cause damage to insulation when the valve is used or serviced |
| 3. * - Vendor to Specify |
| 4. GSM - Grams per square meter |

Noted